

PHYSICS — GRADE 10 | PRESSURE

Quick Check

Lesson 2: What is Pressure? Force and Area on Flour

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.1: Pressure | Lesson 2

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A 2 kg brick (weight 20 N) is placed gently on flour, first on its large face and then on its small face. Which statement about the force the brick presses on the flour with is correct?

- A** The force is greater on the small face because it is narrower.
- B** The force is 20 N on both faces because the weight does not change.
- C** The force is greater on the large face because it is bigger.
- D** The force on the small face is zero because the brick is balanced.

The large face of a brick measures 20 cm by 10 cm, so its area is 200 cm². Given that 1 m² = 10,000 cm², what is this area in m²?

A 0.02 m²

B 0.002 m²

C 0.2 m²

D 2 m²

A 2 kg brick ($g = 10 \text{ N/kg}$) rests on its small face, which has an area of 0.005 m^2 . What pressure does it exert on the floor?

A 4,000 Pa

B 1,000 Pa

C 0.1 Pa

D 0.00025 Pa

The same brick of weight 20 N makes the deepest dent in flour when it rests on its small face. What is the best explanation?

- A** The small face makes the brick heavier.
- B** The small face pushes with a bigger force than the large face.
- C** The same 20 N is spread over a smaller area, so the pressure is greater.
- D** Force and pressure are the same thing, so the dent depends only on the force.

A force of 30 N acts at right angles on a surface of area 0.5 m². What is the pressure on the surface?

A 0.017 Pa

B 30 Pa

C 60 Pa

D 15 Pa

Which statement correctly defines pressure and its unit?

- A** Pressure is the force multiplied by the area, measured in pascals.
- B** Pressure is the total force on a surface, measured in newtons.
- C** Pressure is the mass of an object, measured in kilograms.
- D** Pressure is the force acting at right angles per unit area, measured in pascals where $1 \text{ Pa} = 1 \text{ N/m}^2$.

Mama Wanjiru pushes with the same 40 N force on a sharp knife and on a blunt knife. The sharp knife cuts a tomato easily. Using what you learnt from the brick on flour, what is the best explanation?

- A** The pressure is the same for both knives because the force is the same.
- B** The sharp knife has a bigger force on the tomato than the blunt knife.
- C** The sharp knife has a larger contact area, so it cuts better.
- D** The sharp edge has a much smaller contact area, so the same force gives a greater pressure.

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).